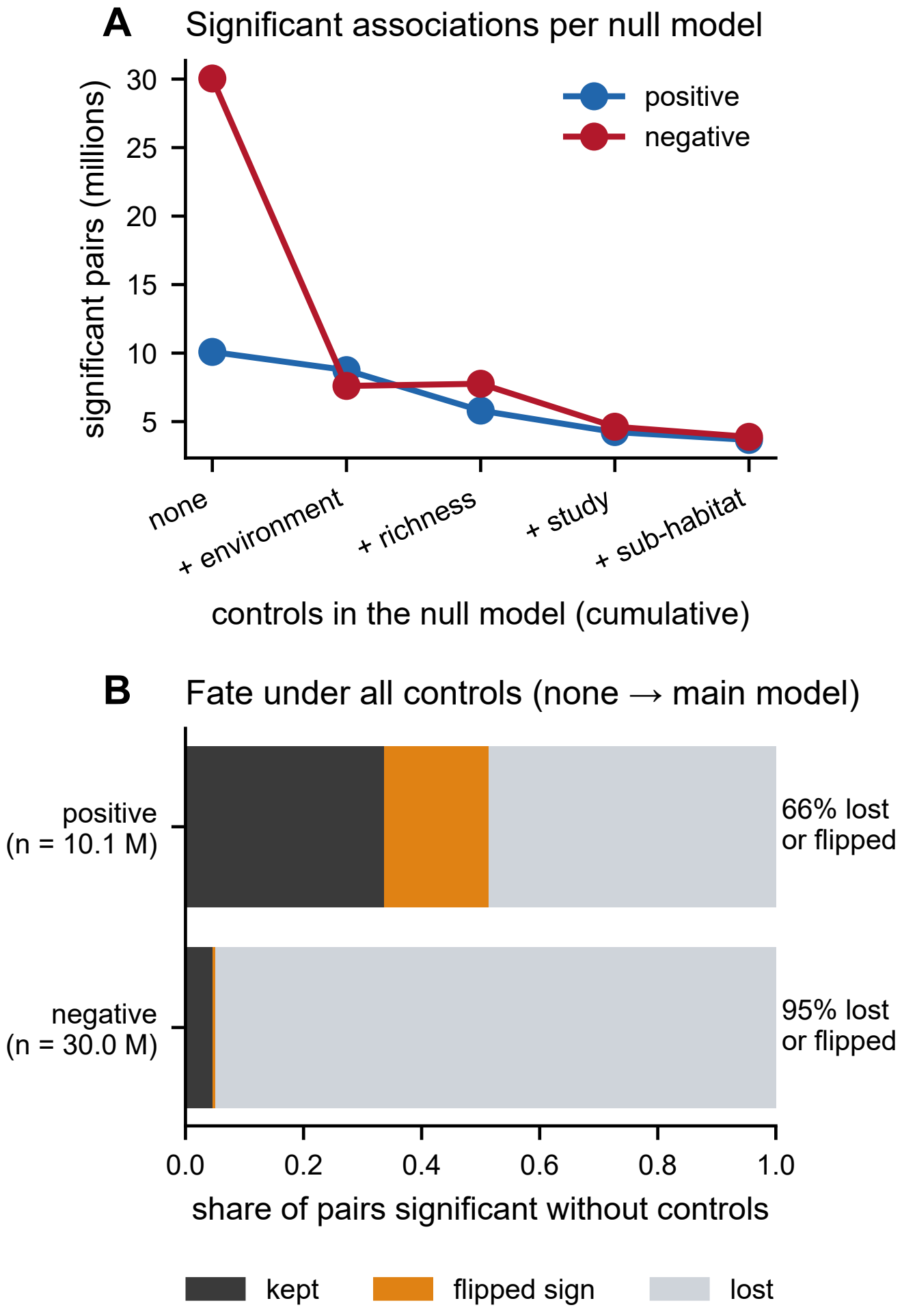
A
Significant associations per null model
significant pairs (millions)
30
25
20
15
10
5
positive
negative
none
+ environment
+ richness
+ study
+ sub-habitat
controls in the null model (cumulative)
B
Fate under all controls (none → main model)
positive
(n = 10.1 M)
negative
(n = 30.0 M)
66% lost
or flipped
95% lost
or flipped
0.0
0.2
0.4
0.6
0.8
1.0
share of pairs significant without controls
kept
flipped sign
lost